Microsoft

"When This Baby Gets Up to 88 Miles Per Hour, You're Gonna See Some Serious Shit."

To 30 Years in the Future From 30 Years in the Past

Daniel Gustafsson

PGConf.EU 2026 - Valencia, Spain


THE TOP OF MIND
LOVE WITHOUT LIMITS




THE TOP OF MIND
(7) TO FLUSHING-MAIN ST




TAKE THE


THE
TOP
OF
MIND

FOR $595, YOU GET WHAT NOBODY ELSE CAN GIVE YOU FOR TWICE THE PRICE.

Even at twice the price, you won't find the power of a Commodore 64™ in any personal computer: The Commodore 64 has a built-in memory of 64K.

That fact alone would have sent computer critics and analysts such as Shearson/American Express to the typewriter for the kind of praise you read on the cover.

But there's more. As a quick read here will tell you.



CPU

MOS 6510 @ 0.985Mhz

8-bit data, 16-bit address

RAM

65 536 bytes

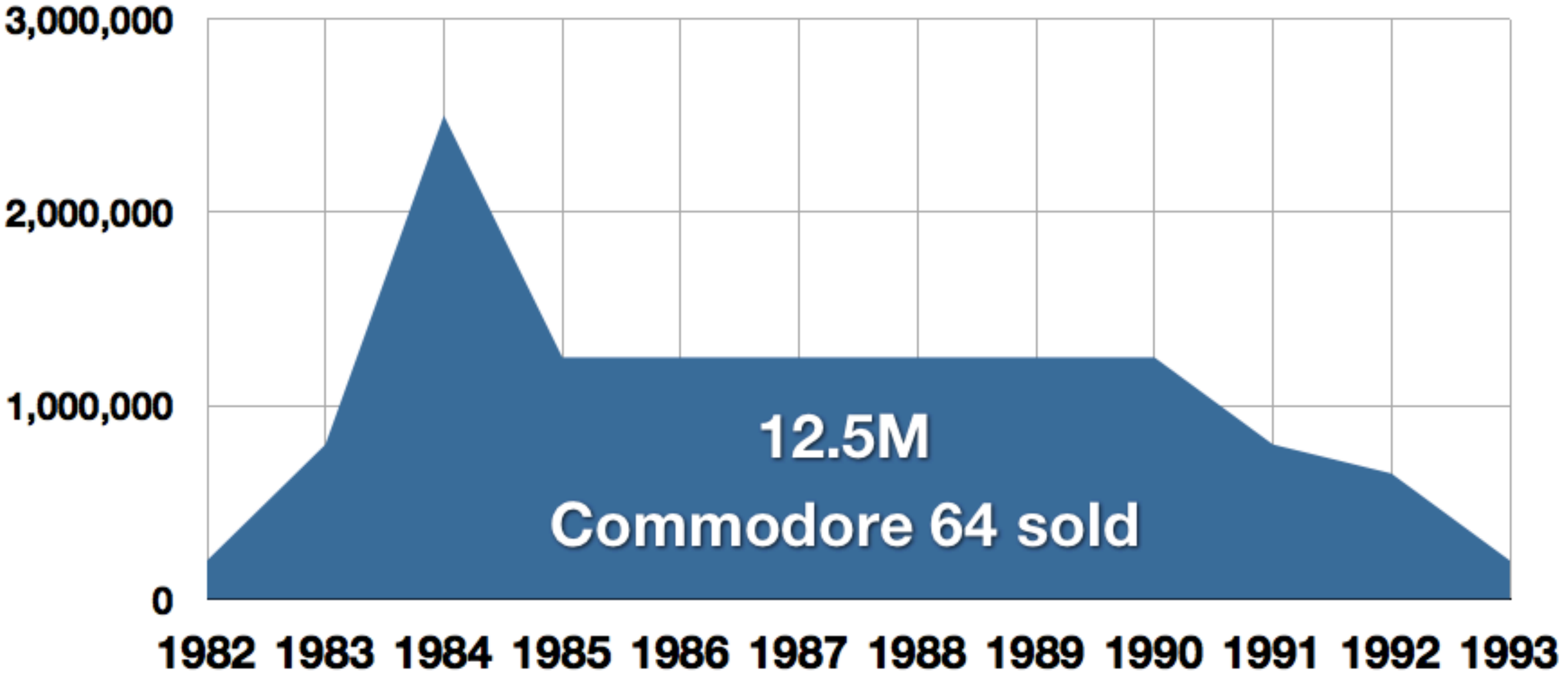
CPU

MOS 6510 @ 1.023Mhz

8-bit data, 16-bit address

RAM

65 536 bytes


3,000,000
2,000,000
1,000,000
0
12.5M
Commodore 64 sold
1982 1983 1984 1985 1986 1987 1988 1989 1990 1991 1992 1993

THE DESIGN OF POSTGRES

Michael Stonebraker and Lawrence A. Rowe

Department of Electrical Engineering
and Computer Sciences
University of California
Berkeley, CA 94720

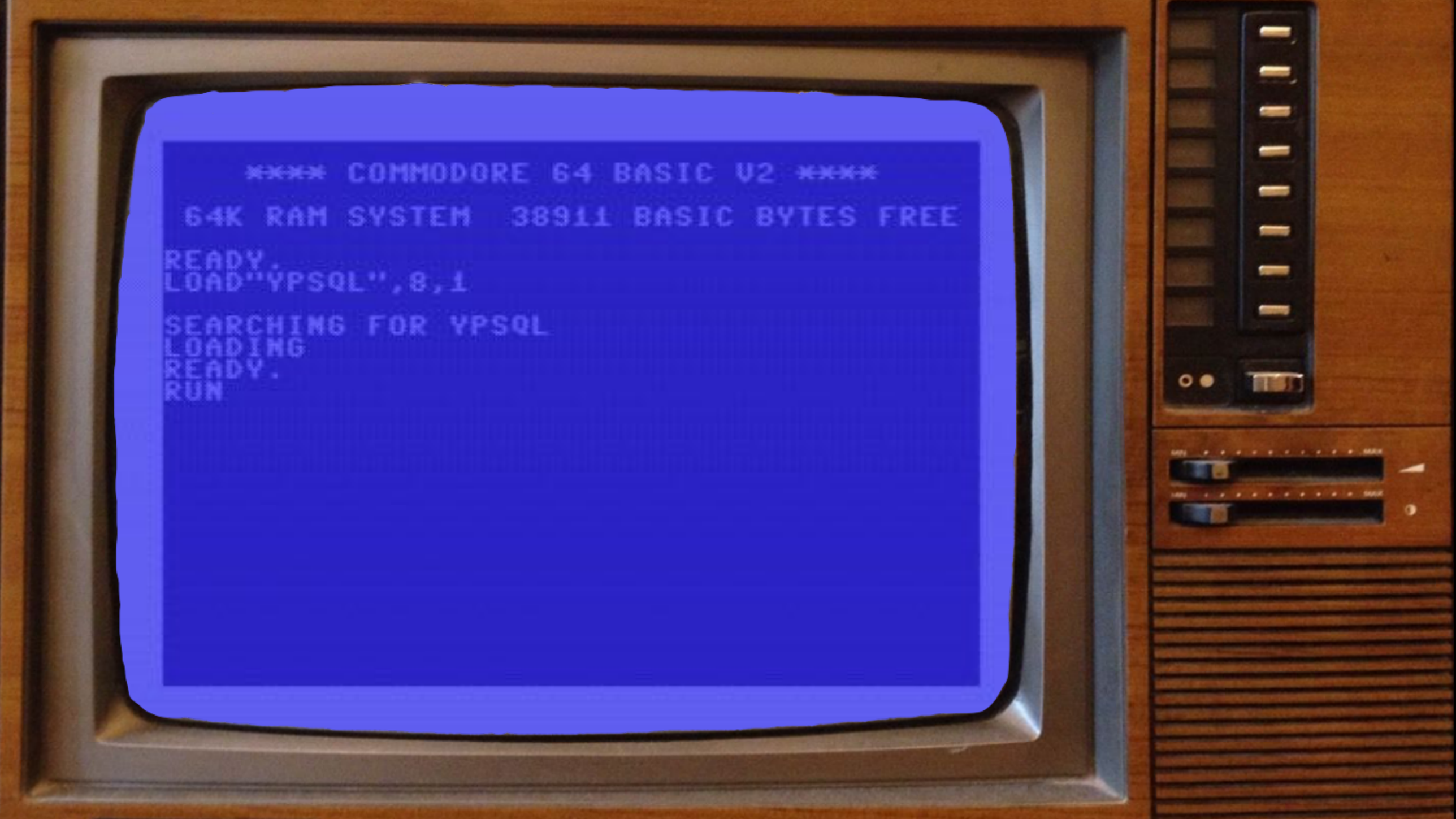
Abstract

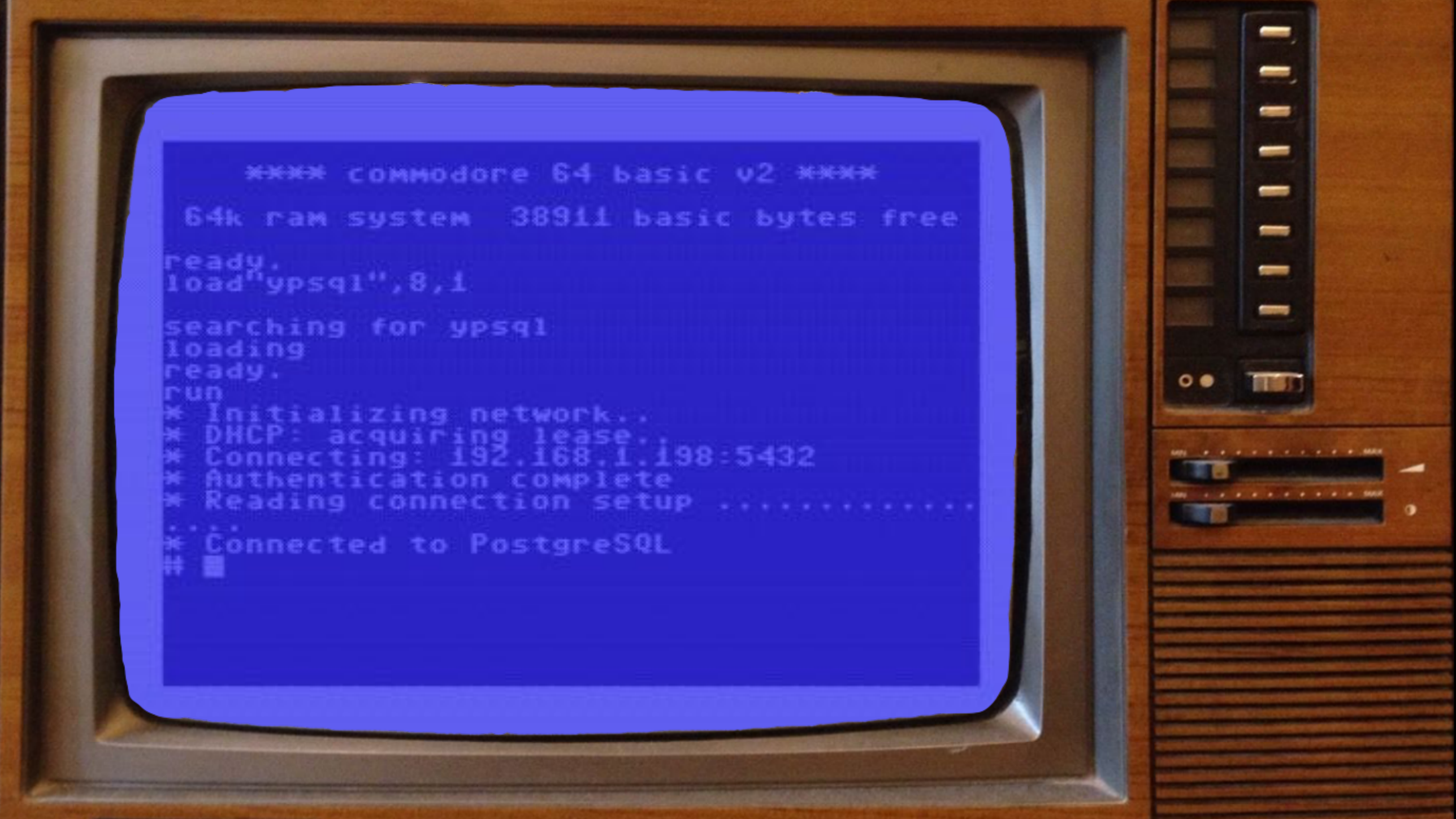
This paper presents the preliminary design of a new database management system, called POSTGRES, that is the successor to the INGRES relational database system. The main design

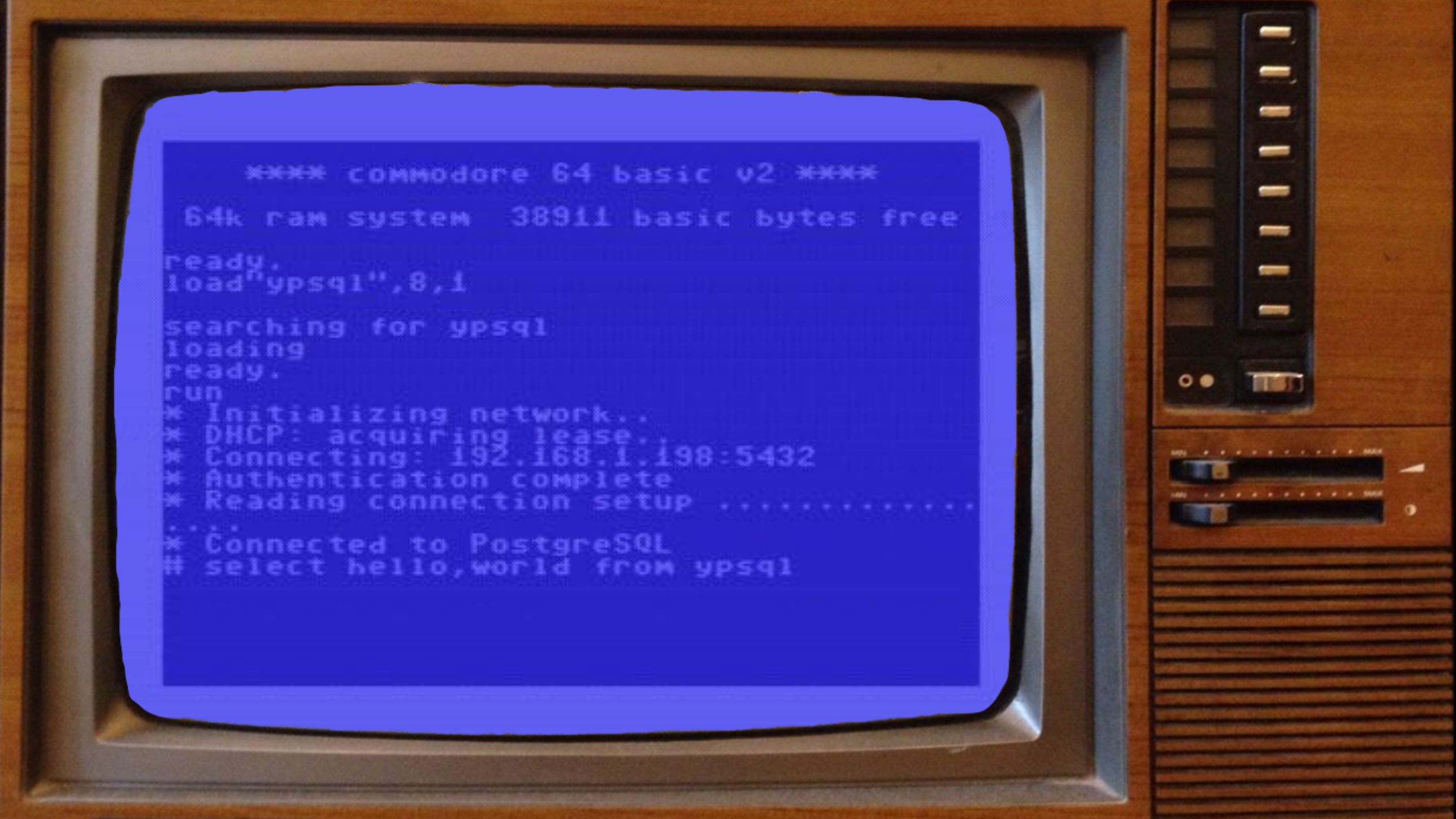
commodore
Highlight All
Match Case
Match Diacritics
Whole Words
Ph

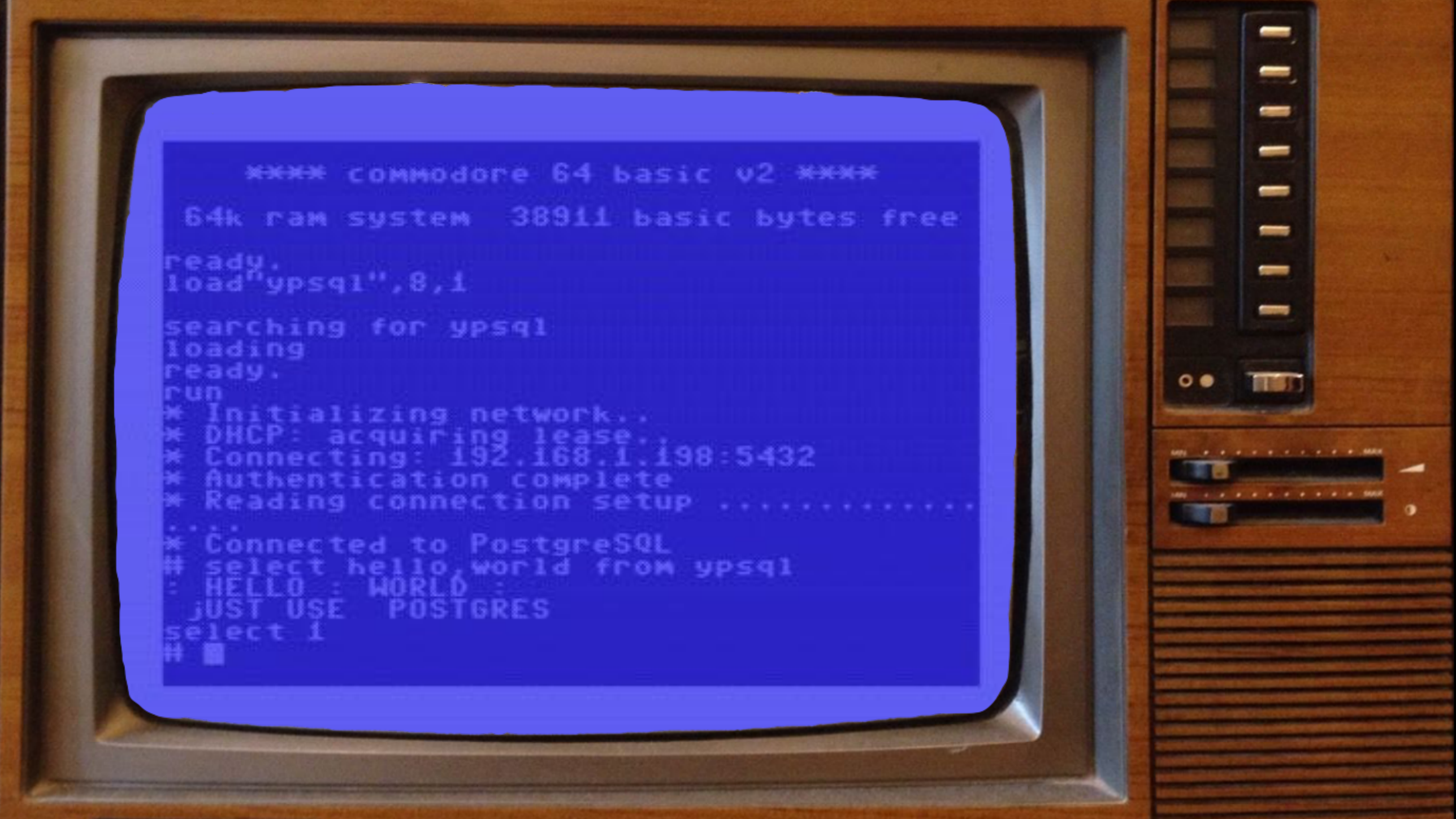
YellowedPlasticSQL

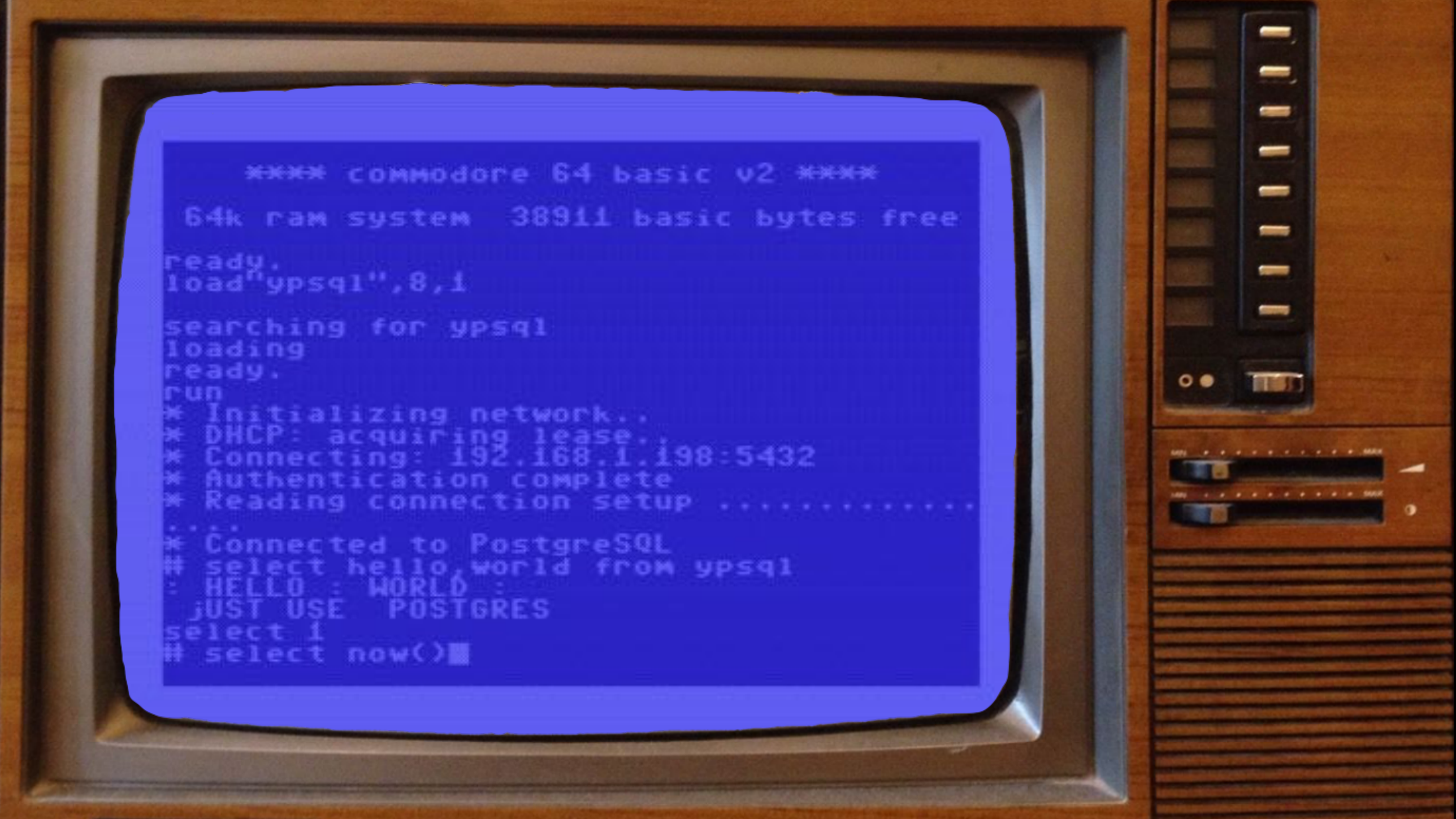


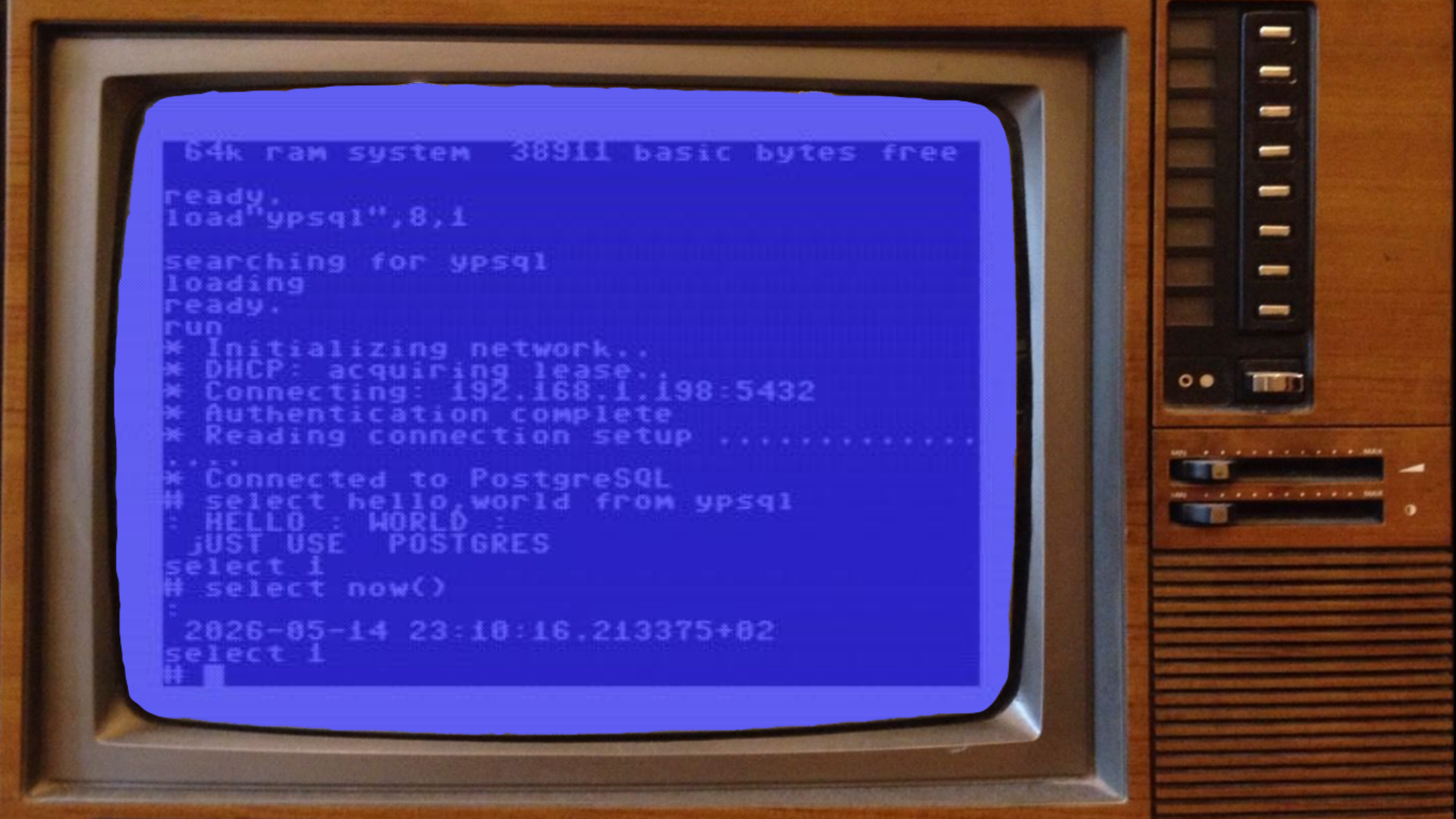

**** COMMODORE 64 BASIC V2 ****
64K RAM SYSTEM 38911 BASIC BYTES FREE
READY.
LOAD"YPSQL",8,1
SEARCHING FOR YPSQL
LOADING
READY.
RUN

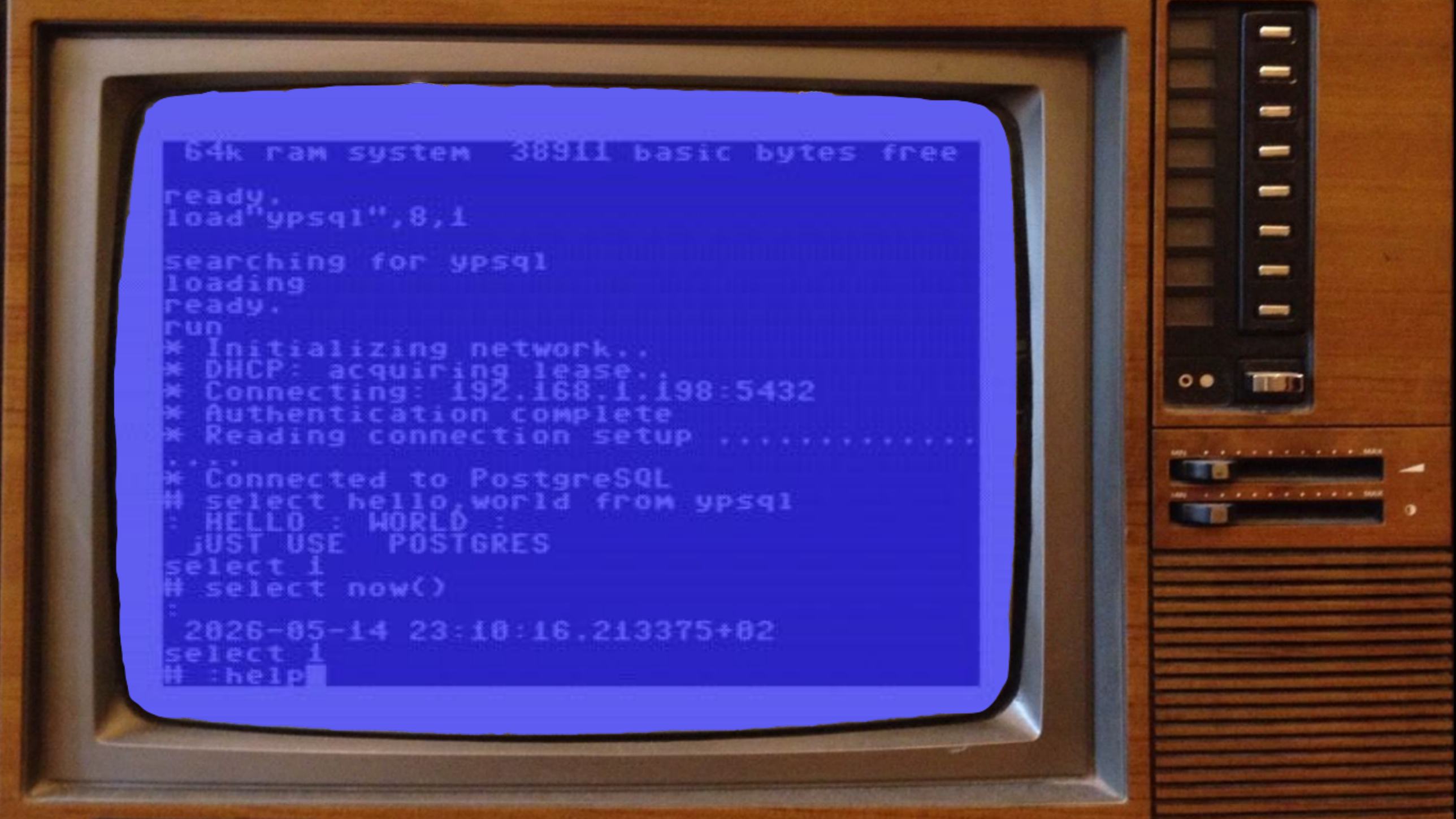

**** commodore 64 basic v2 ****
64k ram system 38911 basic bytes free
ready.
load"ypsql",8,1
searching for ypsql
loading
ready.
run
* Initializing network..
* DHCP: acquiring lease..
* Connecting: 192.168.1.198:5432
* Authentication complete
* Reading connection setup ....
....
* Connected to PostgreSQL
#

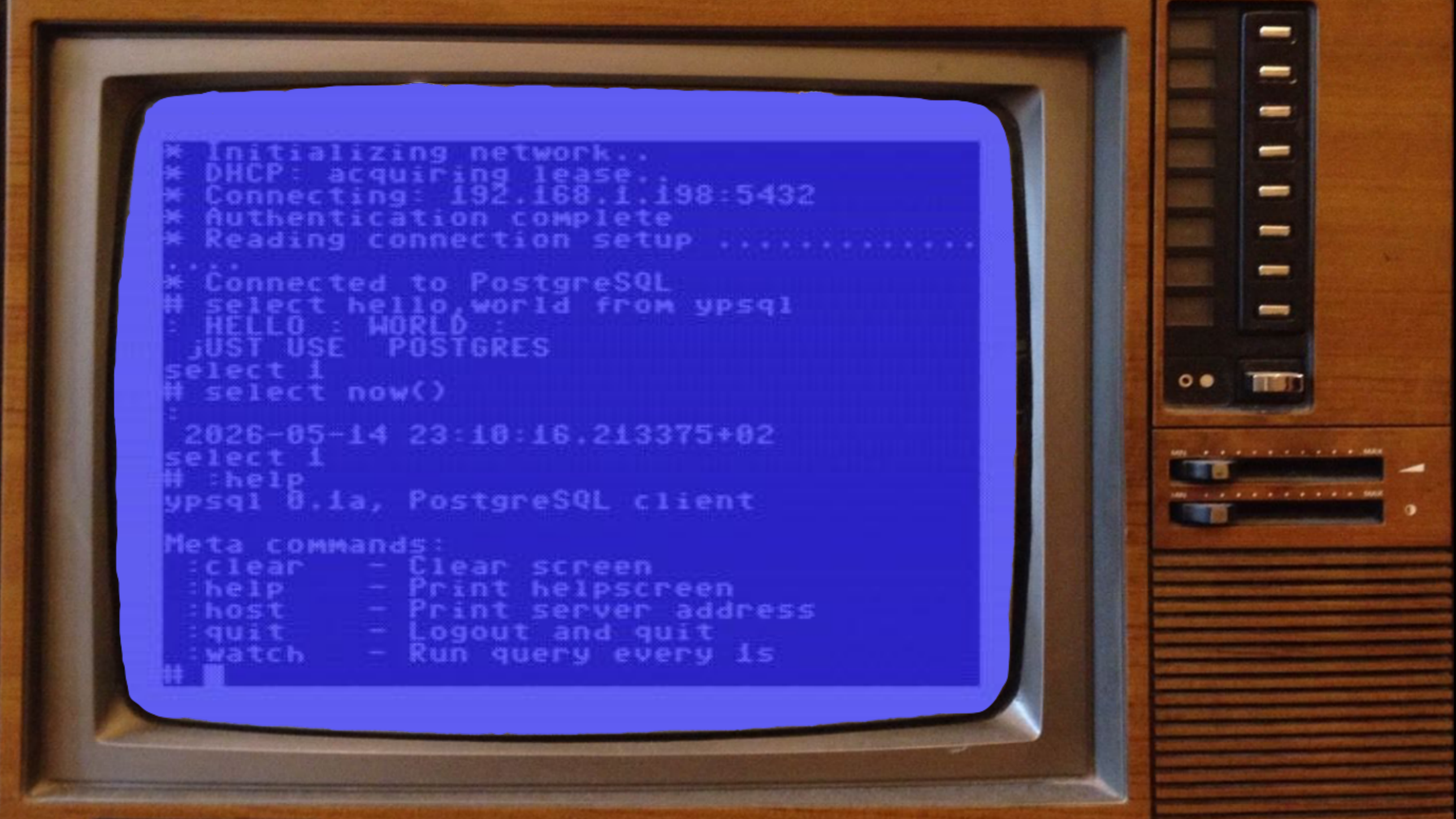

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....
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select hello,world from ypsql

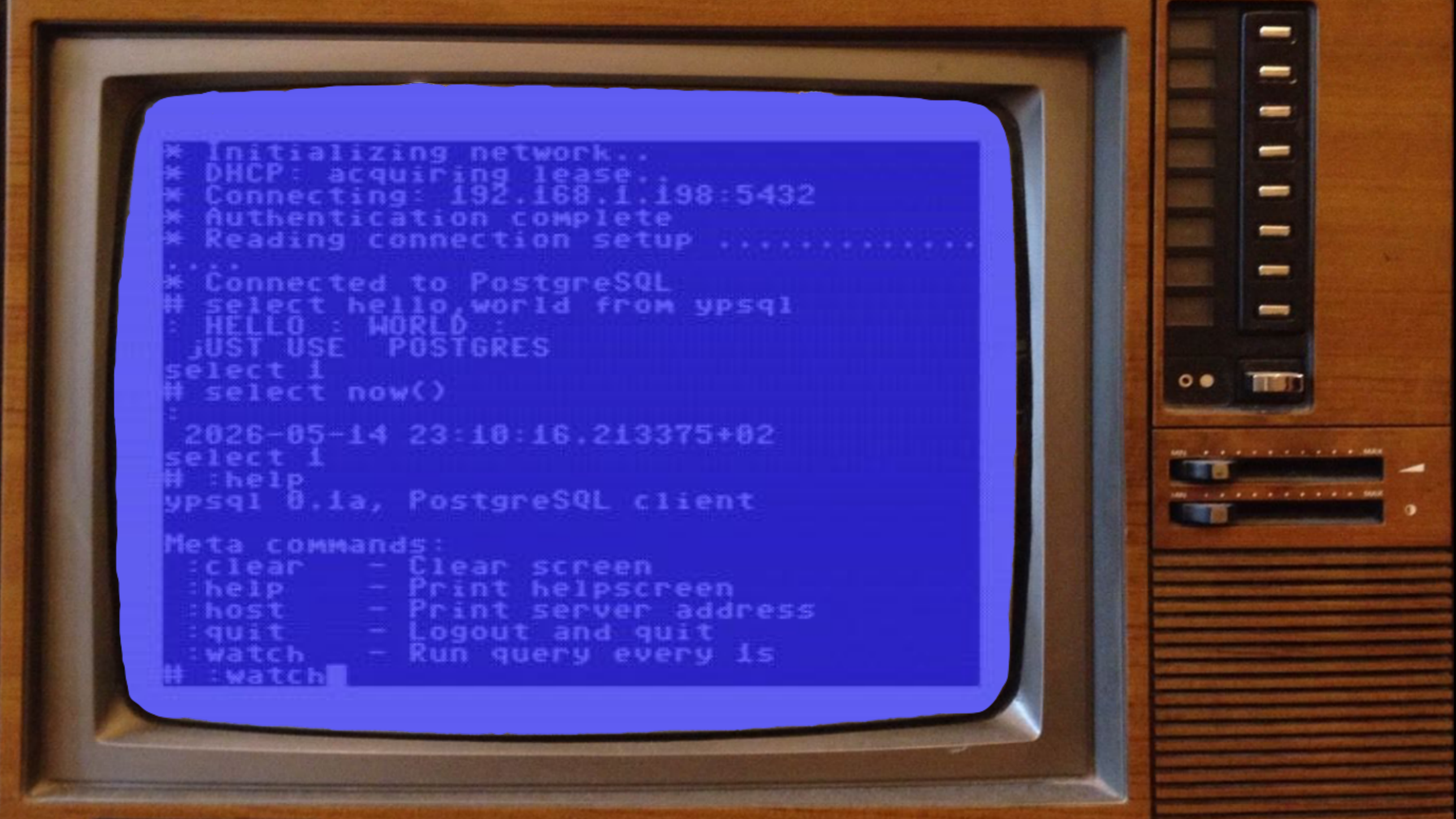

**** commodore 64 basic v2 ****
64k ram system 38911 basic bytes free
ready.
load"ypsql",8,i
searching for ypsql
loading
ready.
run
* Initializing network..
* DHCP: acquiring lease..
* Connecting: 192.168.1.198:5432
* Authentication complete
* Reading connection setup ....
....
* Connected to PostgreSQL
select hello,world from ypsql
: HELLO : WORLD :
: JUST USE POSTGRES
select i
#

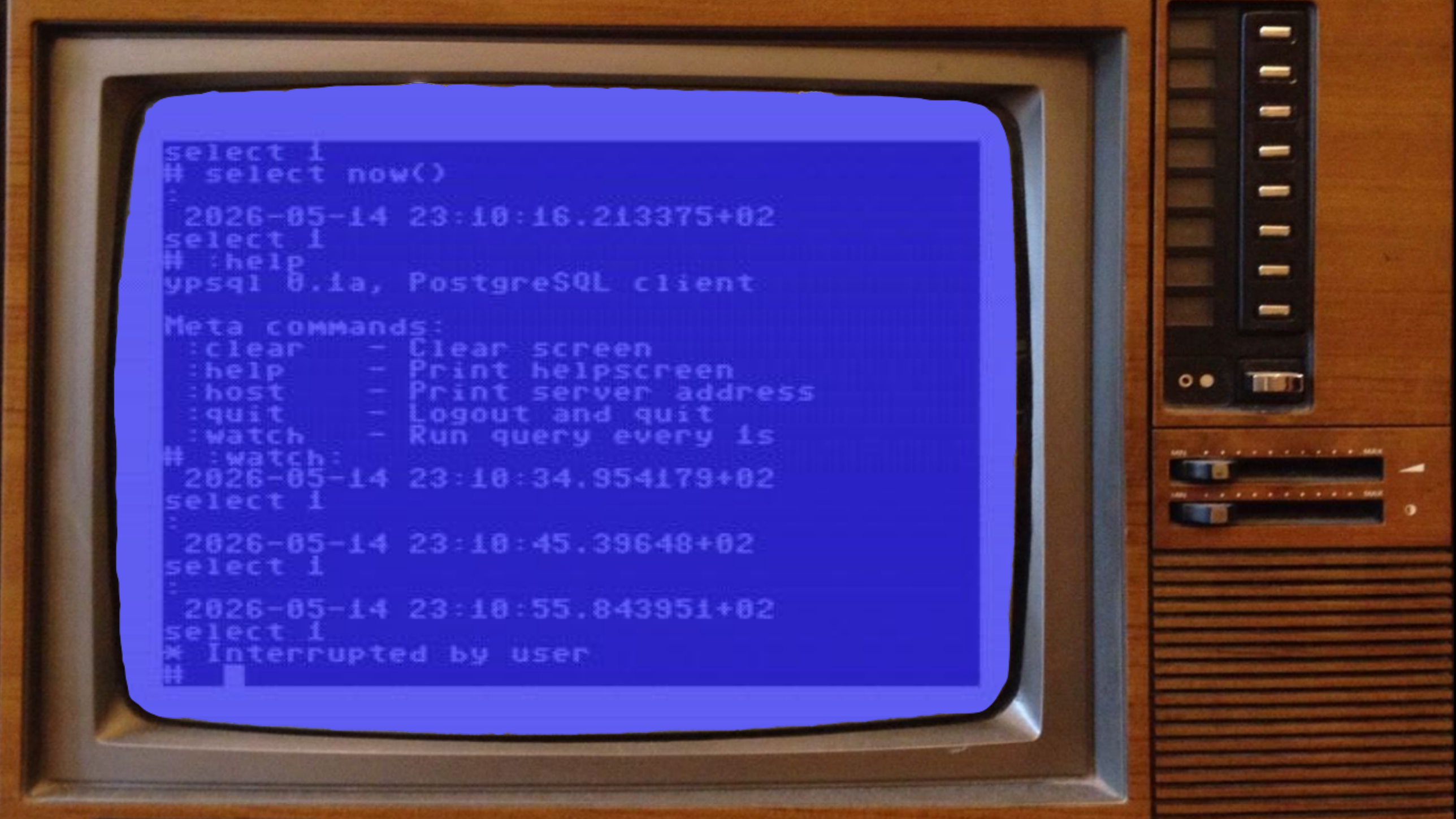

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* Reading connection setup ....
....
* Connected to PostgreSQL
select hello,world from ypsql
: HELLO : WORLD :
: JUST USE POSTGRES
select 1
select now()


64k ram system 38911 basic bytes free
ready.
load"ypsql",8,1
searching for ypsql
loading
ready.
run
* Initializing network..
* DHCP: acquiring lease..
* Connecting: 192.168.1.198:5432
* Authentication complete
* Reading connection setup ....
....
* Connected to PostgreSQL
select hello,world from ypsql
: HELLO : WORLD :
JUST USE POSTGRES
select 1
select now()
:
2026-05-14 23:10:16.213375+02
select 1
#


64k ram system 38911 basic bytes free
ready.
load"ypsql",8,1
searching for ypsql
loading
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run
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* Connected to PostgreSQL
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JUST USE POSTGRES
select 1
select now()
:
2026-05-14 23:10:16.213375+02
select 1
:help


* Initializing network..
* DHCP: acquiring lease..
* Connecting: 192.168.1.198:5432
* Authentication complete
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select hello,world from ypsql
: HELLO : WORLD :
JUST USE POSTGRES
select 1
select now()
:
2026-05-14 23:10:16.213375+02
select 1
:help
ypsql 0.1a, PostgreSQL client
Meta commands:
:clear - Clear screen
:help - Print helpscreen
:host - Print server address
:quit - Logout and quit
:watch - Run query every 1s
#


* Initializing network...
* DHCP: acquiring lease...
* Connecting: 192.168.1.198:5432
* Authentication complete
* Reading connection setup ....
....
* Connected to PostgreSQL
select hello,world from ypsql
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JUST USE POSTGRES
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select now()
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select 1
select now()
;
 2026-05-14 23:10:16.213375+02
select 1
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ypsql 0.1a, PostgreSQL client
Meta commands:
 :clear   - Clear screen
 :help   - Print helpscreen
 :host   - Print server address
 :quit   - Logout and quit
 :watch   - Run query every 1s
:watch:
 2026-05-14 23:10:34.954179+02
select 1
;
 2026-05-14 23:10:45.39648+02
select 1
;
 2026-05-14 23:10:55.843951+02
select 1
* Interrupted by user
#


PostgreSQL client for 8-bit 6502
Runs on hardware/emulator
Requires an RR-Net cartridge
cc65 compiler, ip65 network stack


PostgreSQL wire protocol
Password auth (working on SCRAM)
No TLS (yet..)
\watch command!


Patches welcome!
https://github.com/danielgustafsson/ypsql


TAKE THE
THE TOP OF MIND
PostgreSQL

Microsoft

Thank you!

daniel@yesql.se

https://github.com/danielgustafsson/ypsql